

Master Thesis Presentation

- Good afternoon, my name is Heykel Khadhraoui. I developed a surface grinding controller for a seven-joint robot.
- Normal pressing and compliant rotation allow contact to move the tool towards alignment. A separate experiment tests how the arm responds to disturbances.

Motivation

- Grinding needs the tool face aligned with the physical surface. The configured normal comes from calibration against the plate, with residual uncertainty.
- With a tilted tool, normal pressing creates a contact moment. Low rotational stiffness lets this moment turn the tool towards alignment.
- Cartesian impedance makes position and rotation compliant. The desired orientation stays fixed while the rotational spring and damper remain active.

Contact demonstration

- The video shows pickup, approach, contact, and the hold before grinding.
- During the hold, I change the surface orientation. Normal pressing continues as contact moments rotate the tool.
- The desired orientation stays fixed. This is a qualitative demonstration of compliant contact.

Overview

- I introduce the motivation and Cartesian controller, then the contact experiments and results.
- The separate null-space controller, experiment and results follow together. I finish with the conclusion and future work.

Cartesian pose

- The end effector (EE) is the tool or gripper at the end of the arm. Its Cartesian pose describes position and rotation relative to the robot base frame.
- Position uses x , y , and z , while roll, pitch, and yaw illustrate rotation. Together they give six degrees of freedom.
- Forward kinematics calculates the pose from all seven joint angles, with $p = p(q)$ for position. The controller uses a rotation matrix for orientation.

Cartesian impedance controller

- Positional error points from measured to desired position. Rotational error uses a correction angle ϕ and unit axis u .
- The sketch compares measured and desired tool orientations at one position. The axis points out of the page.
- Stiffness produces force and moment. Damping opposes velocity differences.

- Force and moment form a wrench. Its six-by-six stiffness and damping matrices have three-by-three blocks, with zero off-diagonal blocks here.
- Before the point shift, the stiffness and damping matrices are diagonal in the surface frame.

Real-time control

- The robot receives joint torques. We generate the Cartesian contribution from the wrench using the Jacobian transpose:

$$\tau_{\text{cart}} = J^{\top}(q)F$$

- The Jacobian maps joint velocities to the EE's linear and angular velocities. Model compensation and enabled null-space torques complete the command.
- The measured pose feeds back into the control loop. The C++ controller uses libfranka and updates every millisecond.

Translation–rotation coupling

- The tool centre point (TCP) is the controlled reference point attached to the EE. The centre of compliance (CoC) at p_c is a virtual reference for stiffness and damping.
- Moving the CoC changes how position and rotation errors generate force and moment. Its displacement from the TCP is $r_c = p_c - p_{\text{TCP}}$.
- The full matrix equation uses the same force–moment column and position–rotation error column as the Cartesian controller. The $A^{\top}$ and A transformations shift both stiffness and damping to the TCP.
- The underbraces identify the TCP stiffness and damping matrices. Their off-diagonal blocks couple translation and rotation.
- The error transforms as $e_c = Ae$, and the wrench transforms as $F_{\text{TCP}} = A^{\top}F_c$. These point-shift relations use the small-rotation model.

Centre of compliance (CoC)

- Shifting the CoC can change both the commanded force and moment. The shifted controller's force acts as if applied at p_c :
- The added moment can support or oppose alignment. For the same force, reversing the displacement reverses this contribution.
- With the CoC at the TCP, the added contribution vanishes. The rotational spring and damper still produce m_R .

Contact experiment

- The process goes from tool orientation and surface approach to contact establishment, then grinding. These experiments evaluate the five-second contact-establishment phase.
- Low normal stiffness allows compliant pressing, while higher tangential stiffness resists sideways movement. Rotational stiffness is higher about the normal and lower about the surface tangents.

- The position reference advances into the surface to generate pressing force. The desired orientation stays at its entry value, so contact moments drive the rotation.

Angular quantities

- The red arrow is the inward calibrated surface normal. The green and blue arrows show the tool normals calculated from measured EE orientation at contact entry and end.
- Both angle arcs start from the red reference. The entry angle is the measured angular offset, and the angular error is the remaining difference about the first surface tangent.
- Smaller magnitude means less error in this component. Calibration and mount errors affect the estimate of physical alignment.

Normal-force plausibility assessment

- I manually push the tool about 20 millimetres along the surface normal, with the CoC at the TCP. This checks the spring-force prediction with the desired pose fixed.
- The stiffness is 1000 newtons per metre and the measured position-error change is -19.65 millimetres. Their product gives the spring-force prediction:

$$F_{n,qs} = (1000 \text{ N/m}) \cdot (-19.65 \times 10^{-3} \text{ m}) = -19.65 \text{ N}$$

- In the shaded stationary interval, the commanded mean is -19.64 newtons, only 0.05 percent from the calculation. The robot's model estimate is -22.29 newtons, 13.50 percent from the command.

Moment plausibility assessment

- I manually rotate the tool about surface tangent one while keeping the normal push. This checks the spring-moment prediction with the desired pose fixed and CoC at the TCP.
- The measured rotational error changes by 6.57° relative to the loaded baseline. With a stiffness of 15 newton metres per radian, converting this change to radians gives:

$$M_{t1,qs} = (15 \text{ N m/rad}) \cdot (6.57 \cdot \frac{\pi}{180} \text{ rad}) \approx 1.72 \text{ N m}$$

- In the shaded stationary interval, the commanded mean matches the calculated 1.72 newton metres. The robot's model estimate is 1.85 newton metres, 7.75 percent from the command.

Commanded and estimated wrench

- A separate contact-establishment trial compares commanded force and moment with the robot's model estimates. The CoC is at the TCP.
- During the final stationary interval, mean differences are 2.32 percent for force and 4.87 percent for moment. The curves differ more during motion.

Baseline angular error

- With the CoC at the TCP, entry offsets of 0.69° , $+9.31^\circ$, and -9.41° gave angular errors of 1.65° , 1.75° , and 1.41° .
- The error magnitude decreased for both larger entry tilts, and the negative entry error changed sign. The nominal-zero condition increased from 0.69° to 1.65° .
- The bars show three-trial means, with one sample standard deviation. These are first-tangent components relative to the calibrated surface, calculated from recorded endpoints at 0.01° precision.

Effect of rotational stiffness

- Increasing rotational stiffness from 5 to 50 newton metres per radian increased the angular error from 1.75° to 6.58° . The stiffer rotational spring allowed less contact-induced correction.

Effect of CoC position

- The plot covers minus 80 to plus 80 millimetres, with three repetitions per setting. For positive entry tilt, the smallest tested angular error was 1.17° at plus 80 millimetres. At minus 80 millimetres, 9.35° remained.
- For negative entry tilt, the smallest angular error magnitude was 0.94° at plus 10 millimetres. At minus 80 millimetres, the error component crossed zero and ended at $+2.23^\circ$.
- CoC displacement changes the remaining error component and can reverse its direction. Greater rotation therefore does not always give a smaller angular error.

Angular error and interaction wrench

- In these traces, the error enters and stays within 0.1° of its final value after about 2.5 seconds at plus 40 millimetres. At the TCP this takes 3.6 seconds, about 1.1 seconds longer.
- Final estimated normal forces are similar, at about -83 , -79 , and -78 newtons.
- The interaction moment reverses with CoC displacement. These forces and moments come from the robot's model estimate.

Null-space controller

- The contact results are complete. The separate null-space study concerns the arm's redundant joint motion. At full Jacobian rank, seven joints leave one null-space direction with zero instantaneous EE motion.
- Damping reduces null-space motion by opposing redundant joint velocity.
- Conditioning chooses between two local null-space directions and drives towards the one with the larger minimum singular value. It can initiate joint motion. At a singularity, the EE loses an instantaneous motion direction. Near it, some EE motions require very large joint velocities. A larger minimum singular value indicates better local conditioning under the same Jacobian convention.

Null-space experiment

- The desired position and orientation stay fixed while joint torques apply a repeatable disturbance. This is equivalent to a virtual 20-newton point force on link three.
- We compare no null-space torque, damping, and two conditioning settings, with three trials each. Conditioning is active before and during the disturbance.
- We compare cumulative joint motion, minimum singular value, and net joint motion.

Cumulative joint motion

- Cumulative joint motion, E_N , adds projected joint movement in both directions. These degrees describe arm motion, not rotation of the tool.
- Damping reduces it from 7.60° to 5.69° , about 25.1 percent. Conditioning gives 0.29° at $k_\sigma = 1.5$ and 1.69° at $k_\sigma = 2$.
- Curves show three-trial means over the displayed four-second interval. Shading shows one sample standard deviation.

Net joint motion over time

- This plot retains movement direction. Motion back towards the start reduces the net joint motion.
- The seven projected joint velocities are integrated without taking their magnitude. Each trial uses the same reference direction as the comparison of net joint motion. Positive motion follows the disturbance response without null-space torque.
- The left plot shows all four settings. The right plot enlarges the conditioning curves, in the same degree units. Both show three-trial means and one sample standard deviation.
- Conditioning produces movement away and back towards the starting value. These are projected joint motion values, not tool rotations.

Joint motion over time

- These curves show the measured angle of joint one relative to its value at disturbance onset. Joint one receives the largest equivalent disturbance torque in every trial.
- The left panel shows all four settings over four seconds. Each curve is an individual trial, with markers at the original samples taken at about 20 hertz.
- The right panel enlarges the first second for both conditioning settings. At the larger torque magnitude of 2 newton metres, all three trials show repeated motion reversals. The curves at 1.5 newton metres increase more smoothly.
- Subtracting the starting angle preserves both directions of joint movement. Conditioning was already active before the disturbance, so the settings can begin at different arm configurations.

Joint motion: one trial per setting

- This view shows repetition one for each setting, using the same repetition number throughout the comparison.

- The measured angle of joint one is shown relative to its value at disturbance onset. The axes and conditioning close-up match the preceding slide.
- The larger conditioning torque shows repeated motion reversals in the first second. The lower conditioning setting increases more smoothly during this interval.
- This is one trial per setting. The next slide combines the three trials using their mean and sample standard deviation.

Joint motion: mean of three trials

- This view uses the arithmetic mean of the three trials for every setting. Shading shows one sample standard deviation, matching the uncertainty convention of the other result plots.
- The starting angle is subtracted separately in each trial. The three angle changes are then averaged at their common measured times.
- Repeated reversals remain visible at the larger conditioning torque in the first-second close-up. Averaging reduces reversals that occur at different times in the individual trials.
- These curves show joint one. The cumulative and net joint-motion quantities instead use the projected motion of all seven joints. Conditioning was active before and during the disturbance.

Jacobian conditioning

- A smaller minimum singular value means some tool motions require larger joint velocities. Zero means the EE loses an instantaneous motion direction.
- It falls with no null-space torque and with damping. Both conditioning settings keep it nearly constant during the disturbance.
- Curves show three-trial means over the same four-second interval and use the same Jacobian convention. Shading shows one sample standard deviation.

Net joint motion

- Net joint motion measures projected joint movement remaining after opposite motions cancel. It is 7.52° without null-space torque and 5.61° with damping.
- Mean net joint motion values with conditioning are only $1.5 \times 10^{-2}^\circ$ and $-1.1 \times 10^{-2}^\circ$. Conditioning counters the disturbance in the tested direction.
- Cumulative joint motion values are 0.29° and 1.69° . The larger value is consistent with more back-and-forth motion despite similarly small net joint motion.

Conclusion

- Contact reduced angular error for both larger entry tilts while the desired orientation stayed fixed. The error is relative to the calibrated surface and retains calibration and tool-mount uncertainty.
- Higher rotational stiffness increased angular error from 1.75° to 6.58° .
- CoC displacement changes angular error and can carry it through zero. At the TCP, the added moment vanishes while the rotational spring and damper remain active.

- Damping reduced arm motion, and conditioning countered the tested disturbance direction. Nearly zero net joint motion coexisted with back-and-forth motion.
- Future work is a more rigid tool mount and adaptive CoC selection. After alignment, the adaptive CoC should return to the TCP.

Null-space torques and projector

- The kinematic projector selects null-space torque directions. It uses the Jacobian pseudoinverse J^+ .

$$N_\tau = I_7 - J^\top J^{+\top}$$

- Damping and conditioning add to give the total null-space torque.

$$\tau_{\text{null}} = \tau_d + \tau_\sigma$$

- Damping opposes projected joint velocity and is zero at rest. d_{null} is the damping coefficient.

$$\tau_d = -d_{\text{null}} N_\tau \dot{q}$$

- Conditioning probes both directions along the unit null-space vector v_7 . It chooses the direction with the larger minimum singular value.

$$\tau_\sigma = k_\sigma N_\tau (s_\sigma v_7)$$

- The selector s_σ is +1 or -1, and zero within the deadband. k_σ sets the torque magnitude, which can be nonzero at rest.

Disturbance demonstration

- The controller holds the tool pose throughout the video. Without null-space torque, the disturbance moves the arm.
- Damping opposes the joint motion.
- Conditioning actively changes the arm configuration and can initiate motion. The tool stays approximately in place.